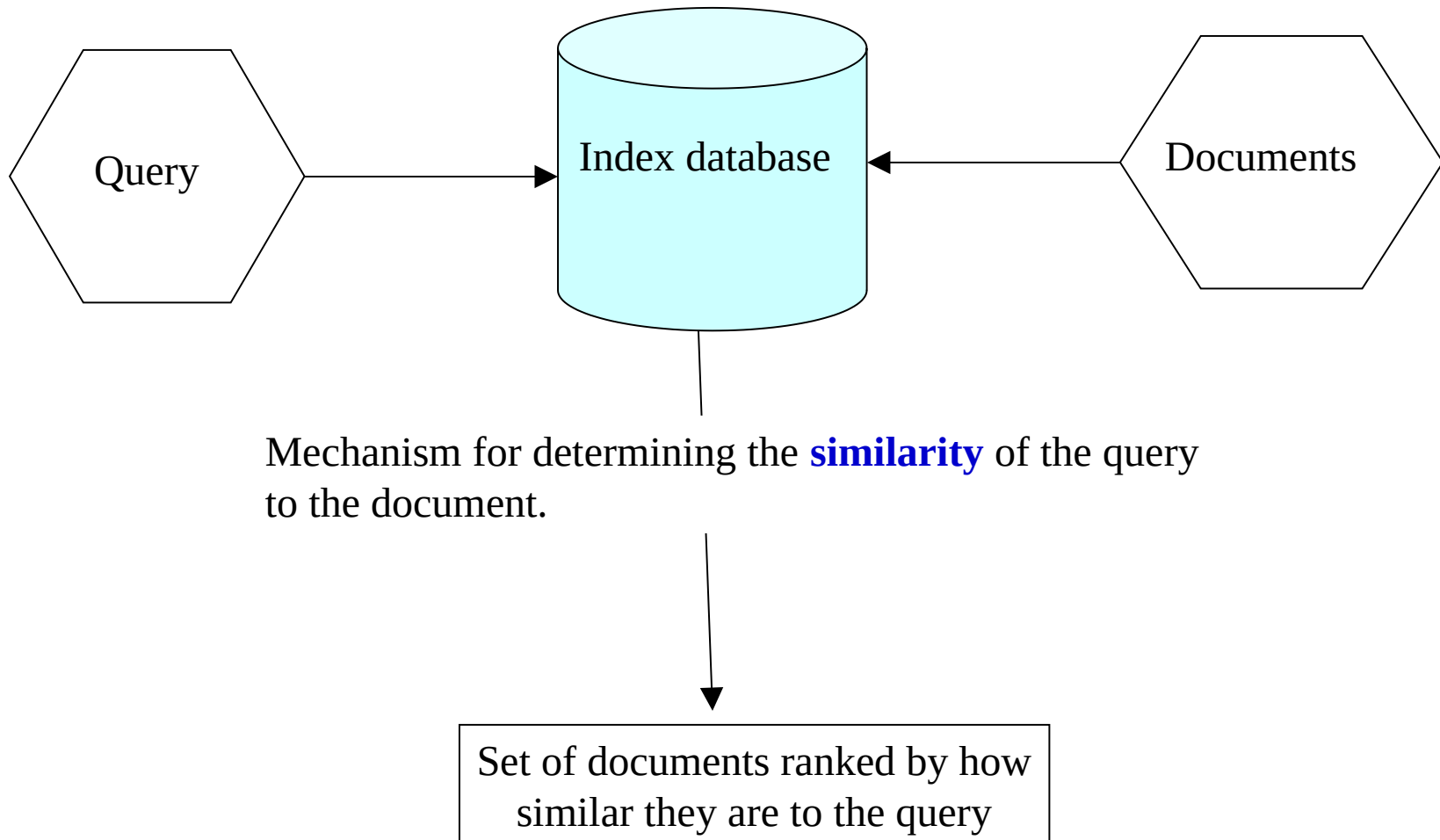


Information Retrieval

Lecture 03

Similarity Ranking Methods

Similarity Ranking Methods



● Weighting: Unnormalized Form of Term Frequency (tf)

document	text	terms
d_1	<i>ant ant bee</i>	<i>ant bee</i>
d_2	<i>dog bee dog hog dog ant dog</i>	<i>ant bee dog hog</i>
d_3	<i>cat gnu dog eel fox</i>	<i>cat dog eel fox gnu</i>

	ant	bee	cat	dog	eel	fox	gnu	hog	<i>length</i>
d_1	2	1							5
d_2	1	1		4				1	19
d_3			1	1	1	1	1		5

Weights: t_{ij} = frequency that term j occurs in document i

● Example (continued)

Similarity of documents in example:

	d_1	d_2	d_3
d_1	1	0.31	0
d_2	0.31	1	0.41
d_3	0	0.41	1

Similarity depends upon the weights given to the terms.

[Note differences in results from example in Lecture 2.]

● Similarity: Weighting by Unnormalized Form of Term Frequency

query		
q	<i>ant dog</i>	
document	text	terms
d_1	<i>ant ant bee</i>	<i>ant bee</i>
d_2	<i>dog bee dog hog dog ant dog</i>	<i>ant bee dog hog</i>
d_3	<i>cat gnu dog eel fox</i>	<i>cat dog eel fox gnu</i>

	ant	bee	cat	dog	eel	fox	gnu	hog	<i>length</i>
q	1			1					$\sqrt{2}$
d_1	2	1							5
d_2	1	1		4				1	19
d_3			1	1	1	1	1		5

● Calculate Ranking

Similarity of query to documents in example:

•	d_1	d_2	d_3
• q	$2/\sqrt{10}$	$5/\sqrt{38}$	$1/\sqrt{10}$
•	0.63	0.81	0.32

- If the query q is searched against this document set, the ranked results are:
- d_2, d_1, d_3

Evaluation

Before we can decide whether one system of weights is **better** than another, we need systematic and repeatable methods to **evaluate** methods of information retrieval.

Methods for Selecting Weights

Empirical

Test a large number of possible weighting schemes with actual data. (*This lecture, based on work of Salton, et al.*)

Model based

Develop a mathematical model of word distribution and derive weighting scheme theoretically. (*Probabilistic model of information retrieval.*)

Weighting

Term Frequency (tf)

Suppose term j appears f_{ij} times in document i . What weighting should be given to a term j ?

Term Frequency: Concept

A term that appears many times within a document is likely to be more important than a term that appears only once.

Normalized Form of Term Frequency: Free-text Document

Length of document

Unnormalized method is to use f_{ij} as the term frequency.

...*but*, in free-text documents, terms are likely to appear more often in long documents. Therefore f_{ij} should be scaled by some variable related to document length.

Term Frequency: Free-text Document

A standard method for free-text documents

Scale f_{ij} relative to the frequency of other terms in the document _{i} . This partially corrects for variations in the length of the documents.

Let $m_i = \max(f_{ij})$ i.e., m_i is the maximum frequency of any term in document i .

Term frequency (tf):

$$tf_{ij} = f_{ij} / m_i$$

Note: There is no special justification for taking this form of term frequency except that it works well in practice and is easy to calculate.

Weighting

Inverse Document Frequency (idf)

Suppose term j appears f_{ij} times in document i . What weighting should be given to a term j ?

Inverse Document Frequency: Concept

By Zipf's Law we know that some terms appear much more often than others across the documents of a corpus.

A term that occurs in only a few documents is likely to be a better discriminator than a term that appears in most or all documents.

Inverse Document Frequency

Suppose there are n documents and that the number of documents in which term j occurs is n_j .

Simple method

We could define document frequency as n_j/n .

A possible method might be to use the inverse, n/n_j , as a weight. This would give greater weight to words that appear in fewer documents.

Inverse Document Frequency

A standard method

The simple method over-emphasizes small differences.
Therefore use a logarithm.

Inverse document frequency (idf):

$$idf_j = \log_2 (n/n_j) + 1 \quad n_j > 0$$

Note: There is no special justification for taking this form of inverse document frequency except that it works well in practice and is easy to calculate.

Example of Inverse Document Frequency

Example

$n = 1,000$ documents

term j	n_j	n/n_j	idf_j
A	100	10.00	4.32
B	500	2.00	2.00
C	900	1.11	1.13
D	1,000	1.00	1.00

Full Weighting:

A Standard Form of tf.idf

Practical experience has demonstrated that weights of the following form perform well in a wide variety of circumstances:

(weight of term j in document i)

= (term frequency) * (inverse document frequency)

A standard tf.idf weighting scheme, for free text documents, is:

$$t_{ij} = tf_{ij} * idf_j$$
$$= (f_{ij} / m_i) * (\log_2 (n/n_j) + 1) \quad \text{when } n_j > 0$$

Vector Similarity Computation with Weights

Documents in a collection are assigned **terms** from a set of n terms

The **term vector space** W is defined as:

if term k does not occur in document d_i , $w_{ik} = 0$

if term k occurs in document d_i , w_{ik} is greater than zero

(w_{ik} is called the **weight** of term k in document d_i)

Similarity between d_i and d_j is defined as:

$$\cos(\mathbf{d}_i, \mathbf{d}_j) = \frac{\sum_{k=1}^n w_{ik} w_{jk}}{|\mathbf{d}_i| |\mathbf{d}_j|}$$

Where $\mathbf{d}_i$ and $\mathbf{d}_j$ are the corresponding weighted term vectors

Structured Text

Structured text

Structured texts, e.g., queries, catalog records or abstracts, have different distribution of terms from free-text. A modified expression for the term frequency is:

$$tf_{ij} = K + (1 - K) * f_{ij} / m_i \quad \text{when } f_{ij} > 0$$

K is a parameter between 0 and 1 that can be tuned for a specific collection.

Structured Text

Query

To weigh terms in the query, Salton and Buckley recommend K equal to 0.5.

However, in practice it is rare for a term to be repeated in a query. Therefore the standard form of tf can be used, i.e., with $K = 0$ and $m = 1$.

Discussion of Similarity

The choice of similarity measure is widely used and works well on a wide range of documents, but has no theoretical basis.

1. There are several possible measures other than angle between vectors
2. There is a choice of possible definitions of *tf* and *idf*
3. With fielded searching, there are various ways to adjust the weight given to each field.

The definitions on Slide 25 can be considered the standard.